Designing the residential tower for maximum profit

Optimal floor plans for 23, 40 and 56 floors

Report for the real estate developer · Prepared by: Grace Chang · October 2026

1 Summary

We determined, for each of the three tower heights under consideration, the design that maximises your profit while meeting all twelve design restrictions: which floor design to use on every floor, who buys each floor, and the sector of every apartment. All three solutions are proven optimal and were checked apartment by apartment against every restriction.

	23 floors	**40 floors**	**56 floors**
Maximum profit	€17.5 m	€29.6 m	€39.2 m
Profit per floor	€760k	€741k	€700k
Apartments	193	334	464

Key findings.

- **Almost all profit comes from large free-sector apartments sold to private buyers.** Every solution places as many 131 m^2 penthouse floors (design ee) at the top of the tower as the rules allow; each such floor earns €1.74 m–1.77.
- **The rest of the tower exists to make those penthouses allowed.** Because at least 40% social and 40% middle apartments are required, free apartments can be at most 20% of all apartments. The lower floors are therefore filled with many small investor apartments (design aa, 12 per floor), which add apartments cheaply and keep the 20% free quota as large as possible.
- **The sector quotas are by far the most expensive rules.** Without the 40% social or 40% middle requirement, profit would be about one third higher. The minimum average size of middle apartments (50 m^2) costs another 10–14%. Six restrictions cost nothing at all.
- **A taller tower earns more in total, but less per floor**, because the profit per apartment falls as the building gets taller.

2 The design problem

The tower has a fixed number of floors. For every floor you choose one of 11 pre-approved floor designs (aa, ab, . . . , ee), each consisting of two of five floor parts with given apartment sizes (36–131 m^2). Every apartment is assigned to a *sector* (social, middle or free) and sold to a *buyer* (housing corporation, investor or private buyer). The profit of an apartment depends on its size, sector, buyer and the tower height; we used the profit tables you supplied for towers of 69 m, 120 m and 168 m, i.e. 23, 40 and 56 floors of 3 m. The design must satisfy the following restrictions.

Whole tower			
R1	at least 40% of apartments social	R3	social apartments at least 40 m^2 on average
R2	at least 40% of apartments middle	R4	middle apartments at least 50 m^2 on average
R9	at least 70% of apartments sold to investors		
Individual apartments			
R5	corporation social apartments $\geq$ 40 m^2	R7	free apartments $\geq$ 60 m^2
R6	corporation middle apartments $\geq$ 50 m^2	R8	corporations buy no free apartments
Floors			
R10	one buyer per floor	R11	no social and free apartments on the same floor
R12	a floor with a smaller smallest apartment lies lower in the tower		

3 How we found the best design

We formulated the problem as a mathematical optimisation model (a mixed-integer linear program, Appendix A) and solved it with the open-source solver CBC. Instead of deciding floor by floor, the model decides *how many* floors of each kind to build (design, buyer and whether the floor may hold free apartments) and *how many* apartments of each size go to each sector. The floors are then stacked in order of their smallest apartment, which satisfies R12 automatically. This keeps the model small: 143 decision variables and 56 constraints, whatever the height of the tower. Each case is solved to proven optimality in under two seconds, so new profit figures or rules can be evaluated instantly.

To make sure the results are correct we (i) converted every solution into an explicit apartment-by-apartment floor plan and checked all twelve restrictions and the profit on it, and (ii) compared the optimum with a separate, much larger model that describes every floor individually, which found the same maximum profit (Appendix C).

4 The optimal designs

	23 floors	**40 floors**	**56 floors**
Profit	€17.48 m	€29.62 m	€39.19 m
Apartments / floor area	193 / 11 822 m²	334 / 20 568 m²	464 / 28 794 m²
Social apartments (avg. size)	78 (40.0 m²)	134 (40.0 m²)	186 (40.0 m²)
Middle apartments (avg. size)	78 (50.2 m²)	134 (50.0 m²)	186 (50.0 m²)
Free apartments (avg. size)	37 (129 m²)	66 (129 m²)	92 (131 m²)
Investor / private / corp. buyers	150 / 36 / 7	262 / 64 / 8	358 / 92 / 14
Investor share (rule: ≥ 70%)	77.7%	78.4%	77.2%
Profit from free / middle / social	€16.0 m / 1.4 / 0.1	€28.4 m / 1.8 / −0.6	€40.0 m / 1.1 / −1.9
Floor designs used	aa×11, ac, cc, cd, ee×9	aa×16, ab, ac×5, bc, cc, ee×16	aa×24, ac×7, cc, ce, ee×23

Table 1: Optimal design for each tower height. Full floor-by-floor plans are in Appendix B.

All three designs follow the same recipe (Figure 1):

1. **Top: private penthouse floors.** Design ee (four 131 m² apartments) sold to private buyers in the free sector. One such apartment earns about €435–444k; no other combination comes close.
2. **Bottom: small investor apartments.** Design aa (twelve apartments of 36–48 m²) sold to investors as social and middle housing. These apartments earn little or even lose money, but each one raises the number of free apartments allowed by the 20% ceiling.
3. **Middle: a few "balancing" floors.** Social apartments must average 40 m² and middle apartments 50 m², while the aa floors are smaller. A few floors with larger apartments (ac, cc, cd, ce) lift the averages to exactly the minimum. Large social apartments are sold to a housing corporation, because for corporations they lose the least money.

In every case the social and middle shares are at (or just above) 40% and both average-size limits are met with almost no slack: these rules determine the design. The investor share (77–78%) is comfortably above 70% without any effort.

5 Which restrictions cost profit

To show where the profit is lost, we re-optimised each case with one restriction removed at a time (Figure 2). The difference is the cost of that restriction to you.

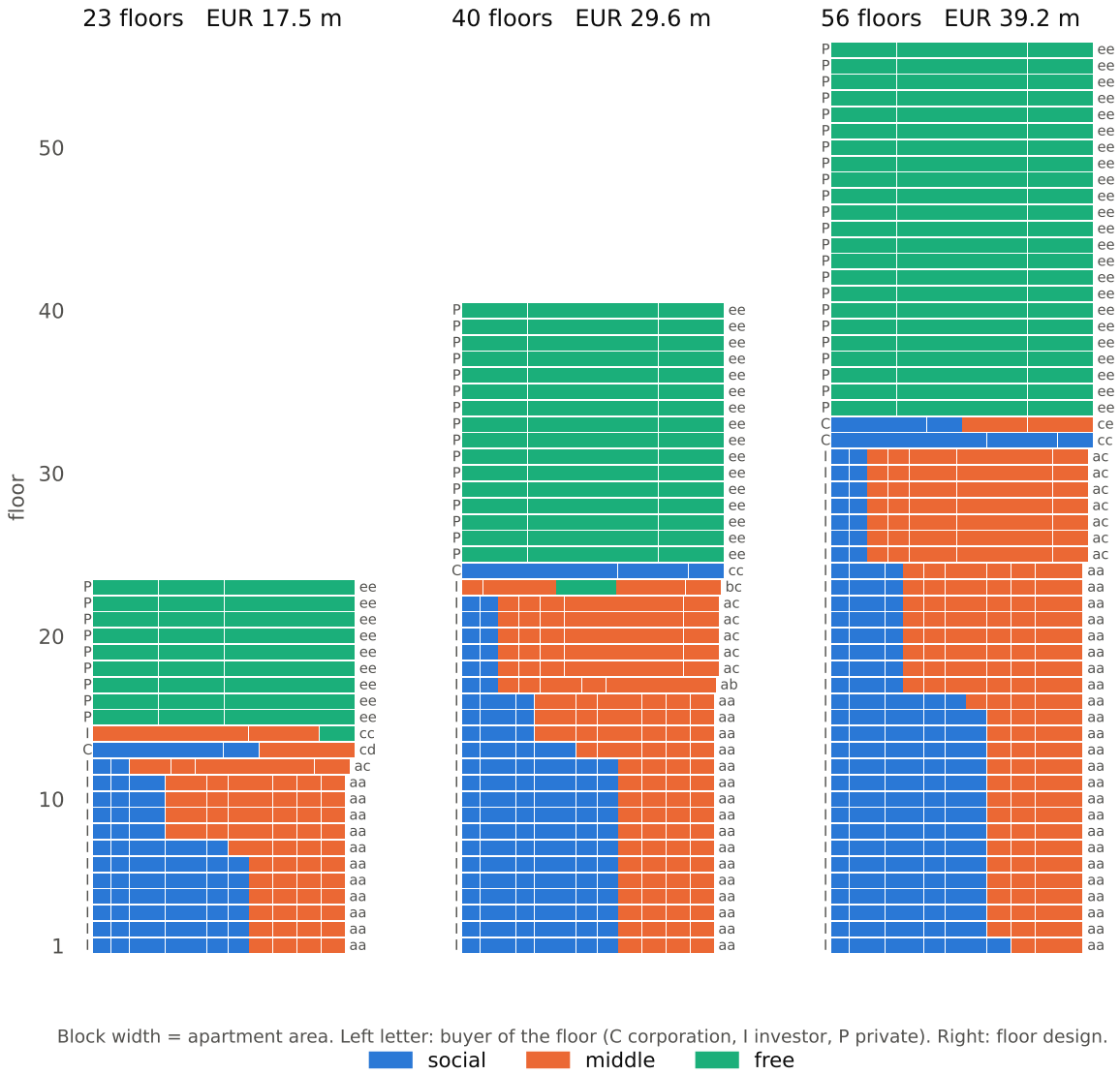


Figure 1: The three optimal towers. Each row is a floor; each block an apartment.

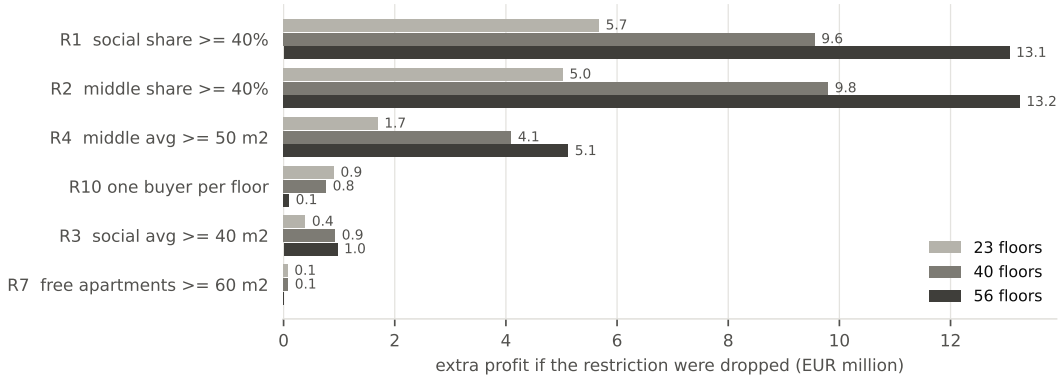


Figure 2: Extra profit if a single restriction were dropped. Restrictions R5, R6, R8, R9, R11 and R12 have no cost in any case and are not shown.

- **R1 and R2 (40% social, 40% middle)** each cost about one third of the profit (€5.0 m–13.2). Any room in these quotas directly turns into more penthouses.
- **R4 (middle at least 50 m^2 on average)** costs €1.7 m–5.1 (10–14%), because it forces larger, loss-making middle apartments. R3, the social counterpart, costs much less (€0.4 m–1.0).

- **R10 (one buyer per floor)** costs up to €0.9 m in the smaller towers; in the 56-floor tower it hardly matters.
- **R5, R6, R8, R9, R11, R12 cost nothing**: the optimal design satisfies them anyway. Agreeing to these rules is free.

6 Recommendations

1. **Build according to the plans in Appendix B** for the chosen height. If the height is still open, the 56-floor tower gives the highest profit (€39.2 m). Each extra floor still pays, but less and less: going from 23 to 40 floors adds €714k per floor, going from 40 to 56 floors €598k per floor. This should be weighed against risks not in the model, such as the sales risk of 92 penthouses.
2. **Focus negotiations with the municipality on the sector quotas and on the middle-size rule.** Every percentage point of extra free housing, or a lower average-size requirement for middle apartments, is worth far more than concessions on the other rules (Figure 2).
3. **Do not spend effort on the investor quota or the social/free mixing rule**; they do not limit the design.
4. **Re-run the model when figures change.** The results depend entirely on the profit tables: the strategy relies on private penthouses being very profitable and small social apartments roughly breaking even. The model takes seconds to re-solve with new data.

Limitations. The profits are taken as given and certain. Other designs with exactly the same profit may exist; we report one optimal design. Within a floor, which individual apartment receives which sector is free to choose, as long as the numbers in Appendix B are kept.

A Mathematical model

Modelling idea. Floor numbers only matter for R12, which only prescribes the *order* of the floors. The model therefore decides how many floors of each type are built, and the order is fixed afterwards (Proposition 2). A floor type is characterised by its design d, its buyer o (R10) and its *mode* t: it may hold social and middle apartments ($t = \text{SM}$) or middle and free apartments ($t = \text{MF}$), which is exactly R11. Within a group (o, t) all apartments of the same area are interchangeable, so only their number per sector is needed (Proposition 1).

Sets.

$\mathcal{D}$	floor designs $\{aa, ab, ac, bb, bc, cc, cd, ce, dd, de, ee\}$
$\mathcal{S}$	apartment areas $\{36, 42, 48, 52, 60, 68, 70, 71, 96, 131\}$ (m^2)
$\mathcal{O}$	buyers $\{\text{corp}, \text{inv}, \text{priv}\}$ $\quad$ $\mathcal{K}$ sectors $\{\text{soc}, \text{mid}, \text{free}\}$
$\mathcal{T}$	floor modes $\{\text{SM}, \text{MF}\}$, with sectors $\mathcal{K}_{\text{SM}} = \{\text{soc}, \text{mid}\}$, $\mathcal{K}_{\text{MF}} = \{\text{mid}, \text{free}\}$
$\mathcal{G}$	admissible floor groups (o, t); $\quad$ Ω admissible apartment assignments (s, o, t, k) (below)

Parameters.

N	number of floors (23, 40 or 56)
n_{ds}	number of apartments of area s on a floor of design d; $\quad n_d = \sum_s n_{ds}$
p_{sko}	profit of an apartment of area s, sector k, buyer o (from the table for height $3N$ m)
α_k	minimum sector share: $\alpha_{\text{soc}} = \alpha_{\text{mid}} = 0.4$ (R1, R2)
A_k	minimum average area: $A_{\text{soc}} = 40$, $A_{\text{mid}} = 50$ (R3, R4)
β	minimum investor share: 0.7 (R9)
$a^{\min}_{ko}$	minimum area per apartment: $a^{\min}_{\text{soc,corp}} = 40$ (R5), $a^{\min}_{\text{mid,corp}} = 50$ (R6), $a^{\min}_{\text{free},o} = 60$ (R7), else 0
Φ	forbidden sector–buyer pairs: $\{(\text{free}, \text{corp})\}$ (R8)

$$\mathcal{G} = \big\{(o,t) \in \mathcal{O} \times \mathcal{T} : t = \text{SM or } (\text{free}, o) \notin \Phi\big\},$$
$$\Omega = \big\{(s,o,t,k) : (o,t) \in \mathcal{G},\ k \in \mathcal{K}_t,\ (k,o) \notin \Phi,\ s \geq a^{\min}_{ko}\big\}.$$

R5–R8 are built into Ω: no variable exists for a forbidden combination.

Decision variables. $y_{dot} \in \mathbb{Z}_{\geq 0}$: number of floors with design d, buyer o and mode t, $(o,t) \in \mathcal{G}$. $x_{sotk} \in \mathbb{Z}_{\geq 0}$: number of apartments of area s in group (o,t) assigned to sector k, $(s,o,t,k) \in \Omega$. With $U = \sum_{d,(o,t)} n_d\, y_{dot}$ (all apartments), $X_k = \sum_{(s,o,t):(s,o,t,k)\in\Omega} x_{sotk}$ and $Q_k = \sum_{(s,o,t):(s,o,t,k)\in\Omega} s\, x_{sotk}$:

$$\max \quad \sum_{(s,o,t,k)\in\Omega} p_{sko}\, x_{sotk} \tag{1}$$
$$\text{s.t.} \quad \sum_{d\in\mathcal{D}} \sum_{(o,t)\in\mathcal{G}} y_{dot} = N \tag{2}$$
$$\sum_{k:(s,o,t,k)\in\Omega} x_{sotk} = \sum_{d\in\mathcal{D}} n_{ds}\, y_{dot} \qquad \forall\, s \in \mathcal{S},\ (o,t) \in \mathcal{G} \tag{3}$$
$$X_k \geq \alpha_k\, U \qquad k \in \{\text{soc}, \text{mid}\} \quad \text{(R1, R2)} \tag{4}$$
$$Q_k \geq A_k\, X_k \qquad k \in \{\text{soc}, \text{mid}\} \quad \text{(R3, R4)} \tag{5}$$
$$\sum_{d,t:(\text{inv},t)\in\mathcal{G}} n_d\, y_{d,\text{inv},t} \geq \beta\, U \qquad \text{(R9)} \tag{6}$$

(3) gives every built apartment exactly one admissible sector; (4) and (5) are the linear forms of $X_k/U \geq \alpha_k$ and $Q_k/X_k \geq A_k$. R10 and R11 hold by construction (buyer and mode belong to the floor), R12 by sorting.

Size. $|\mathcal{D}| \cdot |\mathcal{G}| = 11 \cdot 5 = 55$ variables y and $|\Omega| = 88$ variables x (9+7 for (corp,SM), 10+10 for (inv,SM) and (priv,SM), 10+6 for (inv,MF) and (priv,MF)): 143 integer variables and $1{+}10{\cdot}5{+}5 = 56$ constraints, independent of N. A floor-by-floor model needs at least $N(|\mathcal{D}| + |\mathcal{O}| + |\mathcal{S}||\mathcal{K}|) = 44N$ variables (1 012 to 2 464).

Correctness.

Proposition 1. *Every feasible (y, x) corresponds to a tower satisfying R1–R11 with the same profit, and vice versa.*

($\Rightarrow$) Build y_{dot} floors of each type. By (3), the apartments of area s in group (o, t) can be labelled one by one so that exactly x_{sotk} get sector k; every label is admissible (R5–R8), each floor has one buyer (R10) and one mode (R11), and (4)–(6) are R1–R4, R9. *($\Leftarrow$)* Give each floor of a feasible tower mode MF if it has a free apartment (then it has no social one by R11) and SM otherwise; corporation floors get SM by R8. Counting floors and apartments gives a feasible (y, x). □

Proposition 2. *Let $m_d = \min\{s : n_{ds} > 0\}$. Numbering the floors of an optimal solution such that $m_{d(1)} \le \cdots \le m_{d(N)}$ yields a tower that is optimal under R1–R12.*

Sorting satisfies R12, changes neither the profit nor R1–R11, and adding R12 can only lower the optimum, which the sorted tower attains. □

B Floor plans

Floors are numbered from the ground up. Within a floor, the sectors may be assigned to any apartments of the stated sizes.

23 floors

Floors	Design	Buyer	Apartments (number × m^2)
1–6	aa	Investor	SOCIAL: 4×36, 4×42; MIDDLE: 4×48
7	aa	Investor	SOCIAL: 4×36, 3×42; MIDDLE: 1×42, 4×48
8–11	aa	Investor	SOCIAL: 4×36; MIDDLE: 4×42, 4×48
12	ac	Investor	SOCIAL: 2×36; MIDDLE: 2×42, 2×48, 2×60, 2×71
13	cd	Corporation	SOCIAL: 2×60, 1×70, 2×71; MIDDLE: 2×96
14	cc	Investor	MIDDLE: 4×60, 3×71; FREE: 1×71
15–23	ee	Private	FREE: 4×131

40 floors

Floors	Design	Buyer	Apartments (number × m^2)
1–12	aa	Investor	SOCIAL: 4×36, 4×42; MIDDLE: 4×48
13	aa	Investor	SOCIAL: 4×36, 2×42; MIDDLE: 2×42, 4×48
14–16	aa	Investor	SOCIAL: 4×36; MIDDLE: 4×42, 4×48
17	ab	Investor	SOCIAL: 2×36; MIDDLE: 4×42, 2×48, 2×52, 1×68
18–22	ac	Investor	SOCIAL: 2×36; MIDDLE: 2×42, 2×48, 2×60, 2×71
23	bc	Investor	MIDDLE: 2×42, 2×52, 1×68, 2×71; FREE: 2×60
24	cc	Corporation	SOCIAL: 4×60, 4×71
25–40	ee	Private	FREE: 4×131

56 floors

Floors	Design	Buyer	Apartments (number × m^2)
1	aa	Investor	SOCIAL: 4×36, 4×42, 1×48; MIDDLE: 3×48
2–15	aa	Investor	SOCIAL: 4×36, 4×42; MIDDLE: 4×48
16	aa	Investor	SOCIAL: 4×36, 3×42; MIDDLE: 1×42, 4×48
17–24	aa	Investor	SOCIAL: 4×36; MIDDLE: 4×42, 4×48
25–31	ac	Investor	SOCIAL: 2×36; MIDDLE: 2×42, 2×48, 2×60, 2×71
32	cc	Corporation	SOCIAL: 4×60, 4×71
33	ce	Corporation	SOCIAL: 2×60, 2×71; MIDDLE: 2×131
34–56	ee	Private	FREE: 4×131

C Code and verification

All results in this report are produced by the notebook `housing_model.ipynb` (Python 3.9+, `pulp`, `openpyxl`, `matplotlib`; the CBC solver ships with PuLP). Place the parameter workbook (`Housing_Problem_-_parameter_values.xlsx`) in the same folder and run all cells.

Sections	Content
1–3	parameters, data, index sets $\mathcal{G}$ and Ω (Appendix A)
4	the model, constraints (1)–(6)
5–8	explicit towers sorted for R12, check of all twelve restrictions, floor plans (Appendix B)
9	one-restriction-at-a-time re-optimisation (Section 5)
10–11	summary statistics (Section 1, Table 1) and Figures 1–2
12	floor-by-floor cross-check; switched off by default (`RUN_CROSSCHECK = False`) because it takes 30 minutes per case

Verification. For every case, the explicit floor plan satisfies all twelve restrictions and its apartment-by-apartment profit equals the model optimum. As an independent check, we also built a straightforward floor-by-floor model (1 932 variables and 2 833 constraints for 23 floors, with R12 as explicit ordering constraints) and solved it with two different solvers, CBC and HiGHS, for 30 minutes each. Both found exactly the same maximum profit (€17 483 911) and no better design, but neither could prove optimality within that time, whereas the compact model proves it in about one second.

Data note. The problem description lists floor part b with a 58 m^2 apartment; the parameter workbook uses 68 m^2, which we followed.